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In [1]: import numpy as np

def check_symmetric(A):

    if A.shape[0] != A.shape[1]:
        return False
    return np.array_equal(A, A.T)

A = np.array([[2, 1, -3],
               [1, 2, -3],
               [-3, -3, -3]])

print ("Matrix A: \n", A)
print ("Is a symmetric?", check_symmetric(A))
```

```
Matrix A:
[[ 2  1 -3]
 [ 1  2 -3]
 [-3 -3 -3]]
Is a symmetric? True
```

```
In [ ]:
```